Exercise Sheet for Control Systems

Solution 9: Loop Shaping Design

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S 9.1: Lead Compensator

i) The Bode plot for this transfer function is shown in Figure 1.

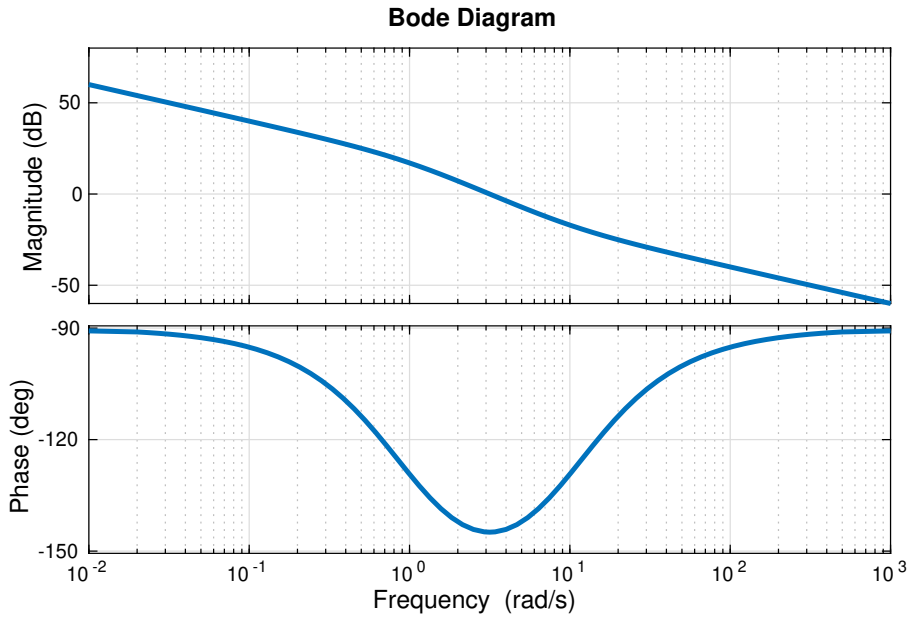

Figure 1 (for S 9.1): Bode plot of the open-loop transfer function $G(s)$.

One can observe that the gain margin is ∞, the phase margin is $35.1°$, and the crossover frequency is 3.16 rad/sec.

ii) The transfer function to the error signal is given by

$$\frac{E(s)}{R(s)} = G_{er}(s) = \frac{1}{1+kG(s)} = \frac{s(s+1)}{s(s+1)+k(s+10)}\,.$$

When applying the final value function for the ramp input, we get

$$\begin{aligned}\lim_{t\to\infty} e(t) &= \lim_{s\to 0} sE(s) = \lim_{s\to 0} sG_{er}(s)R(s) = \lim_{s\to 0} sG_{er}(s)\frac{1}{s^2}\\ &= \lim_{s\to 0}\frac{s+1}{s(s+1)+k(s+10)} = \frac{1}{10k} \overset{!}{\leq} 0.02\,,\end{aligned}$$

which leads to $k \geq 5$. For the following derivations we choose $k = 5$.

iii) The slope of the magnitude plot is -20 dB/decade for frequency $\omega \geq 10$ rad/sec. In order to have a slope -60 dB/decade, we add a filter with two poles at $\omega = 100$ rad/sec. The resulting controller is

$$K(s) = \frac{5}{(1+0.01s)^2}.$$

The resulting Bode plot of the open-loop transfer function $K(s)G(s)$ is shown in Figure 2.

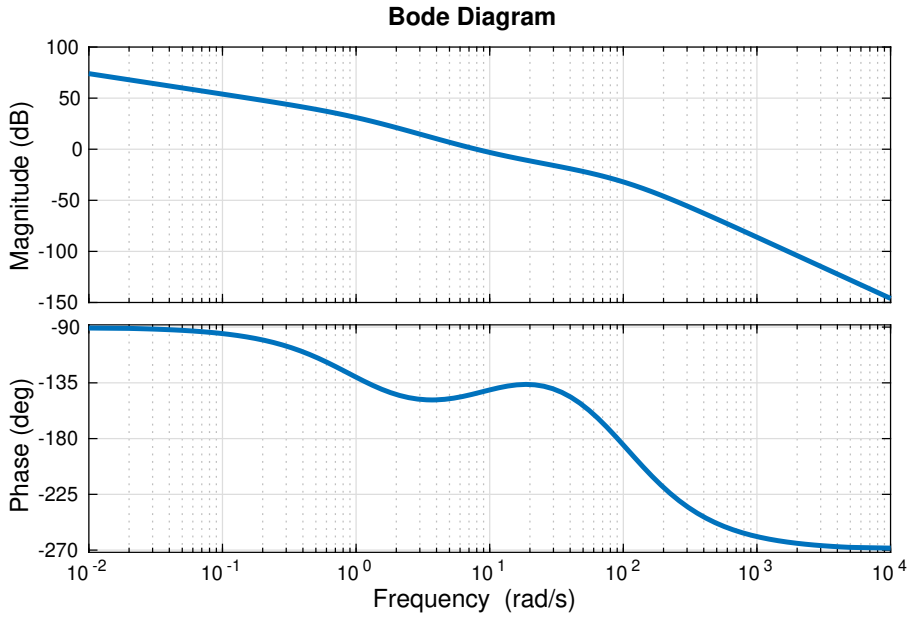

Figure 2 (for S 9.1): Bode plot of the open-loop transfer function $K(s)G(s)$.

iv) Observe from Figure 2 that the phase margin is 36.5° and the gain margin is 30.3 dB. To boost the phase margin, we add a lead compensator in series with the above filter $K(s)$.

Note that we need to add the zero of the compensator just before the crossover frequency to boost the phase margin, and a pole after the crossover frequency to control the gain margin. One possible choice is to add a zero at 5 rad/sec and the pole at 20 rad/sec. The compensator is therefore given by

$$C(s) = \frac{1+0.2s}{(1+0.05s)}.$$

The resulting Bode plot is shown in Figure 3. The phase margin is 77.8° with crossover frequency 14.8 rad/sec, and the gain margin is 21.1 dB.

Note that the resulting compensator is not the unique choice to meet the specifications. You are encouraged to explore other choices of zeros and poles that meet the specifications, and observe how the step response changes with different choices.

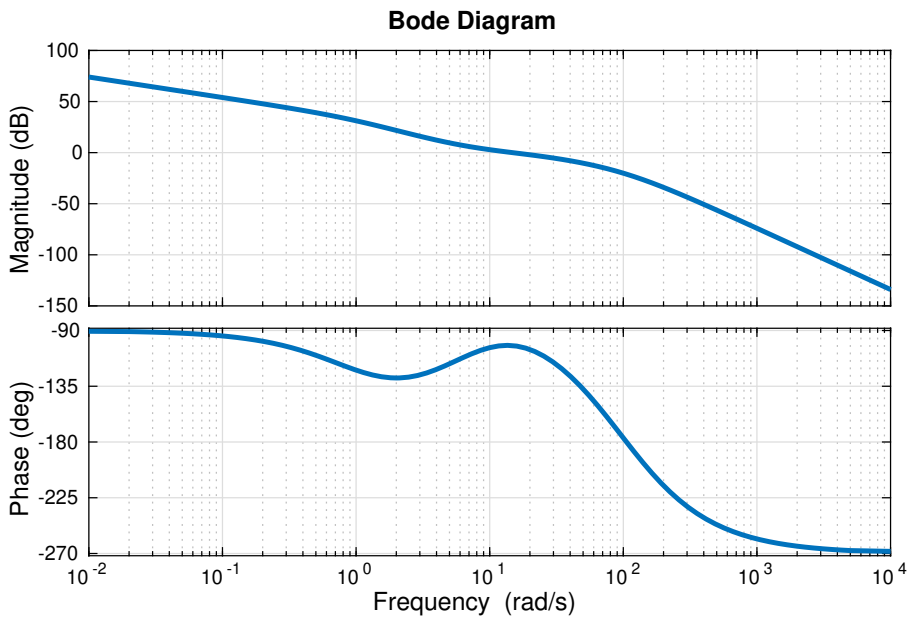

Figure 3 (for S 9.1): Bode plot of the open-loop transfer function $K(s)C(s)G(s)$.

S 9.2: Lead-Lag Compensator

i) The transfer function to the error signal is given as

$$\frac{E(s)}{R(s)} = G_{er}(s) = \frac{1}{1+kG(s)} = \frac{s(1+\frac{s}{10})(1+\frac{s}{200})}{s(1+\frac{s}{10})(1+\frac{s}{200})+k}\,.$$

When applying the final value function we get

$$\begin{aligned}\lim_{t\to\infty} e(t) &= \lim_{s\to 0} sE(s) = \lim_{s\to 0} sG_{er}(s)R(s) = \lim_{s\to 0} sG_{er}(s)\frac{1}{s^2}\\ &= \lim_{s\to 0}\frac{(1+\frac{s}{10})(1+\frac{s}{200})}{s(1+\frac{s}{10})(1+\frac{s}{200})+k} = \frac{1}{k} \overset{!}{\leq} 0.01\,,\end{aligned}$$

which leads to $k \geq 100$. For the following derivations we choose $k = 100$.

ii) The Bode diagram is shown in Figure 4. The actual PM of the uncompensated system is 9.4° (approximately, according to your drafting accuracy). Therefore, the compensator needs to add about $\phi_{max} = 30.6°$ of PM.

The parameter α_1 can be determined by the following formula:

$$\alpha_1 = \frac{1-\sin(\phi_{max})}{1+\sin(\phi_{max})}, \tag{1}$$

For $\phi_{max} = 30.6°$ the result is $\alpha_1 = 0.325$.

We choose $\omega_{\max} = \omega_c$ such that the maximum phase lead occurs at the crossover frequency $\omega_c = 30.7$ rad/s. In other words,

$$\omega_{\max} = 30.7 = \frac{1}{T_1\sqrt{\alpha_1}} \qquad \Longrightarrow \qquad T_1 = 0.057. \tag{2}$$

To keep the crossover frequency of $K_1(s)G(s)$ at the current location, the lead-compensator has to fulfill the following

$$|K_{\text{lead}}(\mathrm{i}\omega)| = 1 \iff k_1\sqrt{\frac{1+(\omega_{\max}T_1)^2}{1+(\alpha_1\omega_{\max}T_1)^2}} = 1 \iff k_1 = \sqrt{\frac{1+(\alpha_1\,\omega_{\max}T_1)^2}{1+(\omega_{\max}T_1)^2}} = 0.57\,.$$

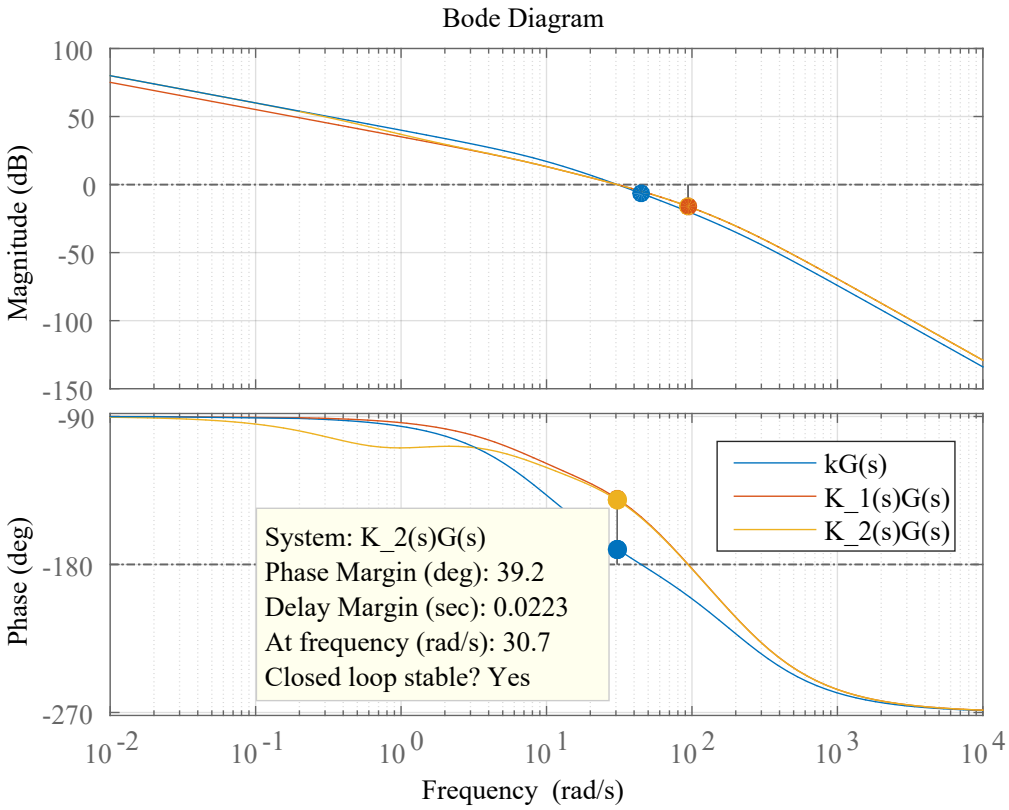

Figure 4 (for S 9.2): Comparison of the Bode plots of the three designs in i), ii) and iii).

iii) With the controller $K_1(s)$ the closed-loop transfer function from the input to the error signal $e(t)$ is given as $G_{er}(s) = \frac{1}{1+K_1(s)G(s)}$

$$\lim_{t\to\infty} e(t) = \lim_{s\to 0} sE(s) = \lim_{s\to 0} sG_{er}(s)\frac{1}{s^2} = \frac{1}{k_1 k} = 0.0175\,,$$

and we can see here the drawback of the lead-compensator, which decreases the gain for $K_1(s)G(s)$ for small frequencies, thus increasing the stationary control error. We can use an additional lag-compensator

$$K_{\text{lag}}(s) = \alpha_2\frac{T_2 s+1}{\alpha_2 T_2 s+1}\,, \quad \alpha_2 > 1$$

to improve the stationary control error. Since with $K_2(s) = kK_{\text{lead}}(s)K_{\text{lag}}(s)$ the stationary control error to a unit ramp reference input is $\lim_{t\to\infty} e(t) = \frac{1}{\alpha_2 k_1 k}$. Thus for a stationary control error of at most 0.01 we need $\alpha_2 \geq \frac{100}{k_1 k} = \frac{1}{k_1} = 1.75$. The T_2 has to be chosen such that the lag-compensator does not influence the phase margin. Therefore the zero introduced by the lag compensator has to be at least one decade smaller than the crossover frequency. This can be achieved for T_2 being large enough such as $T_2 \geq 1$.

iv) Figure 4 shows the comparison of the three designs.

S 9.3: IMC and PID controller

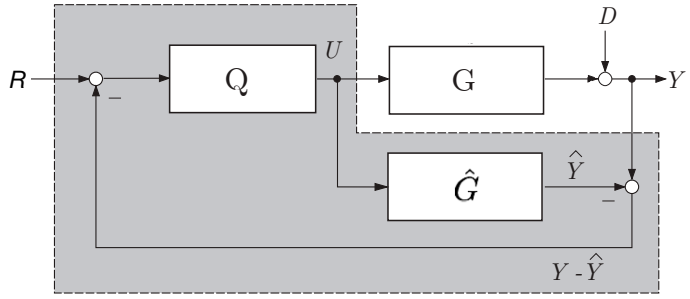

Figure 5 (for S 9.3): IMC control setup

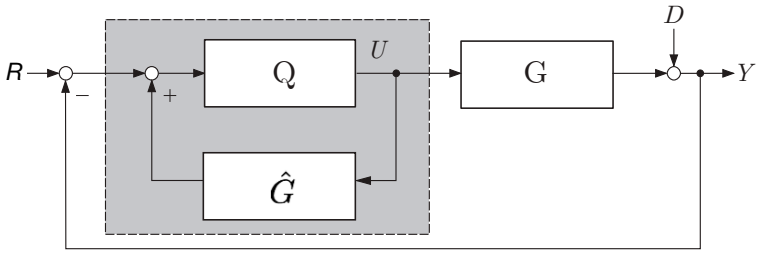

Figure 6 (for S 9.3): Equivalent formulation of the IMC control setup

i) We first consider the block diagram representation shown in Figure 5. Observe that

$$\begin{aligned}
U(s) &= Q(s)[R(s) - Y(s) + \hat{Y}(s)] \\
&= Q(s)[R(s) - D(s) - G(s)U(s) + \hat{G}(s)U(s)] \\
\implies U(s)[1 + Q(s)(G(s) - \hat{G}(s))] &= Q(s)[R(s) - D(s)] \\
\implies U(s) &= \frac{Q(s)}{1 + Q(s)(G(s) - \hat{G}(s))} R(s) - \frac{Q(s)}{1 + Q(s)(G(s) - \hat{G}(s))} D(s). \qquad (3)
\end{aligned}$$

Furthermore, we obtain

$$\begin{aligned}
Y(s) &= D(s) + G(s)U(s) \\
\implies Y(s) &= \frac{G(s)Q(s)}{1 + Q(s)(G(s) - \hat{G}(s))} R(s) + \left(1 - \frac{Q(s)G(s)}{1 + Q(s)(G(s) - \hat{G}(s))}\right) D(s). \qquad (4)
\end{aligned}$$

Thus, dropping the argument s, we obtain

$$\begin{bmatrix} Y \\ U \end{bmatrix} = \begin{bmatrix} \frac{GQ}{1+Q(G-\hat{G})} & \frac{1-Q\hat{G}}{1+Q(G-\hat{G})} \\ \frac{Q}{1+Q(G-\hat{G})} & \frac{-Q}{1+Q(G-\hat{G})} \end{bmatrix} \begin{bmatrix} R \\ D \end{bmatrix}. \qquad (5)$$

We now show that the transfer functions are identical in the representation shown in Figure 6. First observe that

$$\begin{aligned}
U(s) &= Q(s)[\hat{G}U(s) + R(s) - G(s)U(s) - D(s)] \\
\implies U(s)[1 + Q(s)(G(s) - \hat{G}(s))] &= Q(s)[R(s) - D(s)],
\end{aligned}$$

which is equivalent to (3). The calculations for the output follows from identical steps.

ii)
- Every choice except the first is achievable, because $G(s)$ is a transfer function of order 1. Note that we can never get a controlled system of less order than the uncontrolled system with a causal controller.
- In what follows, we choose the desired closed-loop transfer function as $T(s) = \frac{1}{T_f s+1}$. Therefore, we have

$$Q(s) = T(s)G(s)^{-1} = \frac{1}{a}\,\frac{bs+1}{T_f s+1},$$

and the resulting controller is given as

$$K(s) = \frac{Q(s)}{1-\hat{G}(s)Q(s)}\,. \tag{6}$$

Since we assume perfect model knowledge we have $\hat{G}(s) = G(s)$. Thus, the controller in (6) is

$$K(s) = \frac{Q(s)}{1-G(s)Q(s)} = \frac{T(s)G(s)^{-1}}{1-T(s)} = \frac{bs+1}{a(T_f s+1)\left(1-\frac{1}{T_f s+1}\right)} = \frac{b}{aT_f}\left(1+\frac{1}{bs}\right),$$

and equivalent to a PI controller with $k = \frac{b}{aT_f}$ and $T_I = b$.

iii) For the second order system the achievable closed-loop transfer function of lowest order is $T(s) = \frac{1}{(T_f s+1)^2}$ for the same reasoning as before. The resulting controller is given as

$$\begin{aligned} K(s) &= \frac{Q(s)}{1-G(s)Q(s)} = \frac{T(s)G(s)^{-1}}{1-T(s)} = \frac{1}{a}\,\frac{(bs+1)(cs+1)}{(T_f s+1)^2-1} \\ &= \frac{1}{\frac{T_f}{2}s+1}\,\frac{b+c}{2aT_f}\left(1+\frac{1}{(b+c)s}+\frac{bc}{b+c}s\right). \end{aligned}$$

Thus the resulting controller is a filtered PID controller with a filter term $\frac{1}{\frac{T_f}{2}s+1}$ and the gains $k = \frac{b+c}{2aT_f}$, $T_I = b+c$ and $T_D = \frac{bc}{b+c}$.